

2021 JC2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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H2 BIOLOGY

9744/02

Paper 2 Structured Questions

15 September 2021

Candidates answer on Question Paper.
No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

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/100	

This document consists of **27** printed pages and **1** blank page.

Answer **all** questions.

- 1 In 1934, two biologists Davson and Danielli published their suggestion for the structure of the cell surface membrane, as shown in Fig. 1.1.

globular protein

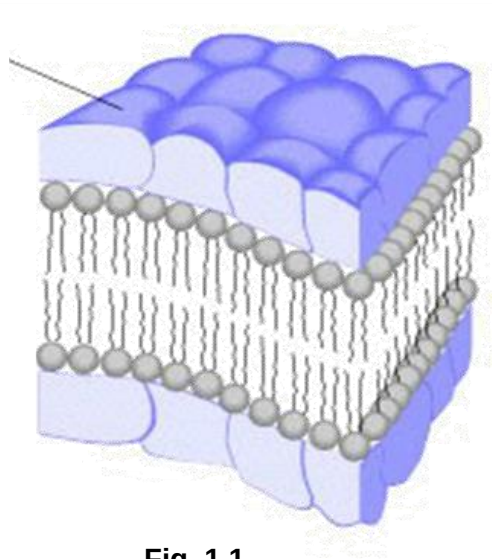


Fig. 1.1

- (a) (i) State one way in which the Davson-Danielli structure is similar to the fluid mosaic model and one way in which it differs from the fluid mosaic model.

Similarity

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Difference

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.....[2]

- (ii) Suggest two problems that the Davson-Danielli structure of the membrane would pose to the functioning of the cell.

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Question 1 continues on page 4

Fig. 1.2 shows the PI3K signalling pathway.

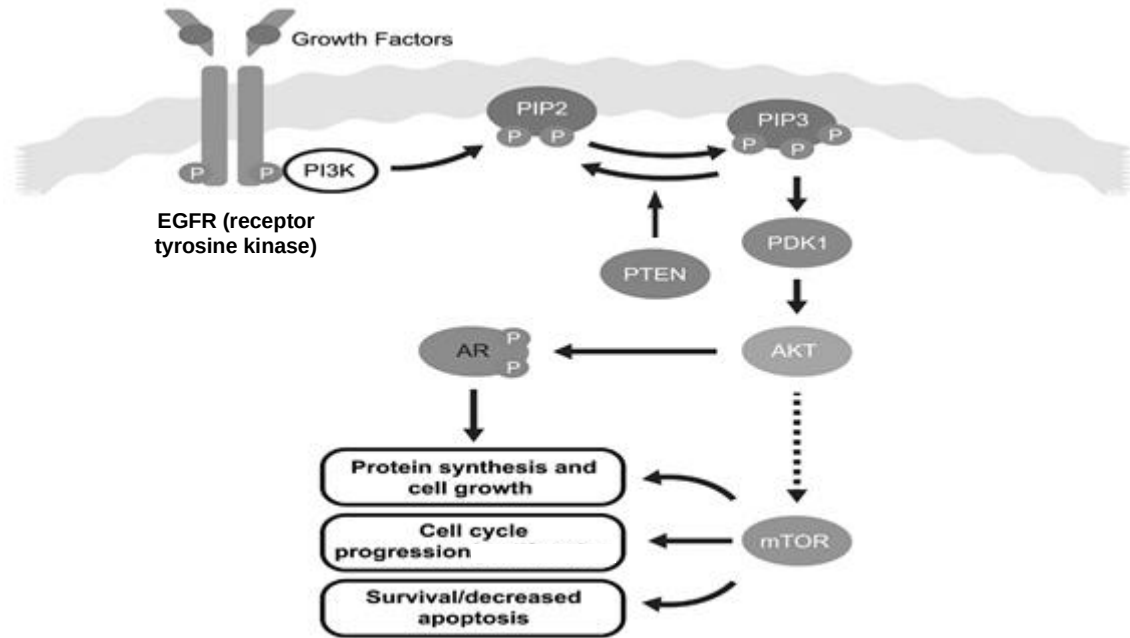


Fig. 1.2

- (b)** Based on biological knowledge and Fig. 1.2, describe what happens after binding of growth factors to EGFR to result in prostate cancer.

.....[4]

- (c) An upregulation of the PI3K signalling pathway is mostly due to the deletion of the tumour suppressor gene, *PTEN*.

In order for the PI3K signaling pathway to be upregulated, both copies of the *PTEN* gene needs to be deleted.

Explain this observation.

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.....[1]

- (d) A 40 year old man decides to do a genetic test. On his report, it was stated that only a homozygous deletion of *PTEN* gene has been detected in the sample of cheek cells that he sent. He concluded that he has oral cancer.

Comment on the validity of his conclusion.

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.....[1]

[Total: 10]

- 2 Collagen is the most abundant protein found in the human body, and is majorly found in connective tissue.

(a) Glycine is a common amino acid found in collagen.

Describe the role of glycine in a collagen molecule.

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Collagen “unwinds” or “shrinks” under heat treatment.

Fig. 2.1 shows how the appearance of a collagen molecule changes after heat treatment, while Fig. 2.2 shows experimental data of how its structure changes with temperature.

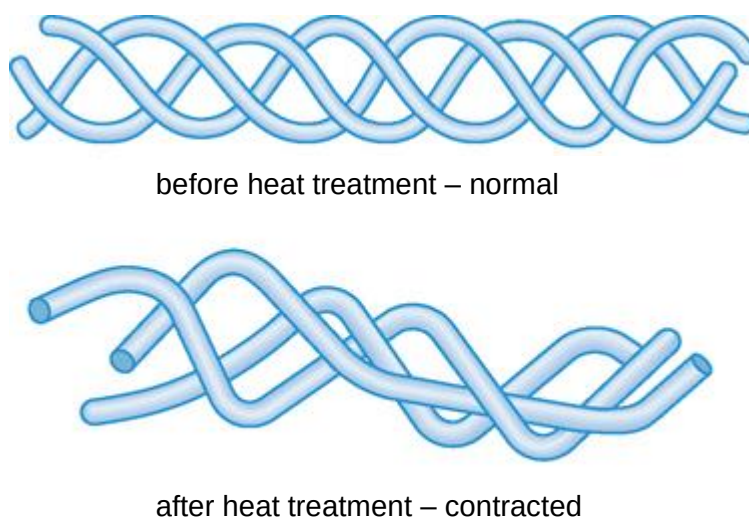


Fig. 2.1

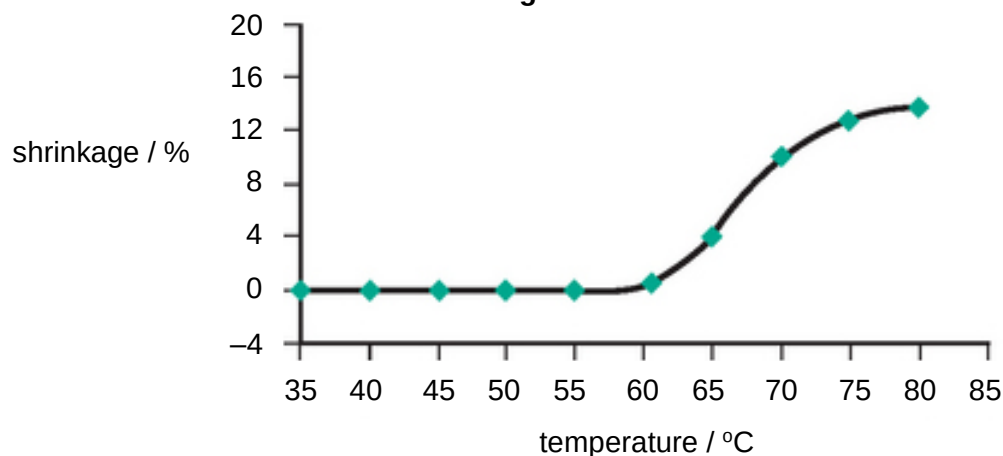


Fig. 2.2

(b) With reference to the information including Fig. 2.1 and Fig. 2.2,

(i) state why collagen is classified as a quaternary protein.

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.....[1]

(ii) describe **and** explain how the structure of a collagen molecule is affected when temperature increases from 35°C to 80°C.

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.....[4]

(c) State how a **named** property of collagen is affected with increasing temperature.

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.....[1]

(d) Suggest why the collagen molecule in Fig. 2.1 does not unwind completely even at high temperatures beyond 80°C.

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.....[1]

[Total: 9]

- 3** In 2008, a woman had her damaged left bronchus replaced by one constructed using her own stem cells. A section of the trachea from a donor was treated in order to remove the cells of the donor, leaving only a framework of collagen. Adult stem cells derived from the woman's own bone marrow were then added to this collagen framework. Samples of these stem cells, together with cells derived from the lining of her trachea, were placed in a bioreactor for four days during which they multiplied to coat the collagen framework. This was used to replace the damaged section of her bronchus. A month later the transplanted tissue had developed its own blood supply.

This was claimed to be the first successful transplant using tissues derived from the patient's own stem cells.

- (a)** Explain the advantages of using a patient's own stem cells to treat a damaged bronchus, as described in the passage, rather than transplanting a complete bronchus.

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- (b)** Stem cells from the human bone marrow that are involved in blood cell formation are described as multipotent, rather than totipotent.

Compare multipotent and totipotent stem cells.

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- (c) Some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In experiments with mice, it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells. There is evidence to suggest that the introduction of the four genes caused an increase in the production of telomerase reverse transcriptase (TERT) in the fully differentiated cells.

Explain how TERT may help to change the cells back into stem cells.

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- (d) Suggest a potential risk with such treatments involving production of TERT in cells.

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[Total: 9]

- 4 *Escherichia coli* (*E. coli*) is a species of bacteria that is commonly found in the gut of humans and warm-blooded animals. While most strains of *E. coli* are harmless, some strains such as Shiga toxin-producing *E. coli* (STEC), can cause severe and potentially fatal foodborne disease. STEC is transmitted to humans primarily through consumption of contaminated foods, such as raw or undercooked ground meat products.

Fig. 4.1 is a micrograph of *E. coli* cells.

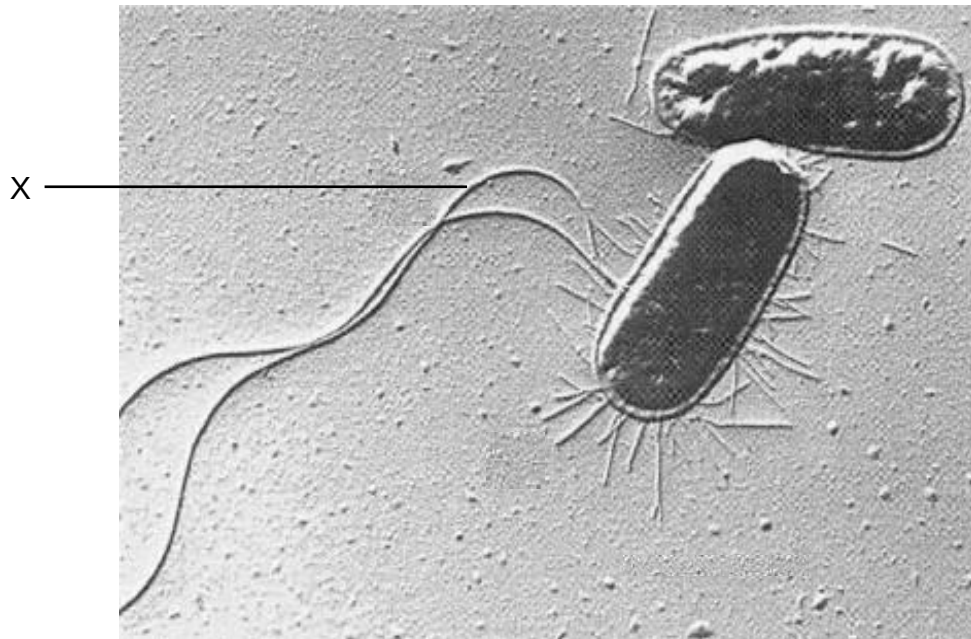


Fig. 4.1

- (a)** Identify structure X.

.....[1]

- (b)** During its life cycle, Shiga toxin-producing *E. coli* (STEC) is exposed to a variety of acidic conditions. Survival in the acidic environments within the hosts is characterised by the induction of several *E. coli* genes upon exposure to acidic conditions.

Fig. 4.2 shows an operon found in STEC involved in the regulation of pH.

1. CadC is a transmembrane protein that is encoded by the *cadC* gene. In the presence of lysine under acidic conditions, CadC is activated.
2. Expression of *cadB* and *cadA* genes is induced.
3. CadA and CadB proteins are produced to help regulate intracellular and extracellular pH. CadA is an enzyme and CadB is a transmembrane protein.
4. Lysine is converted to cadaverine which is a proton acceptor.

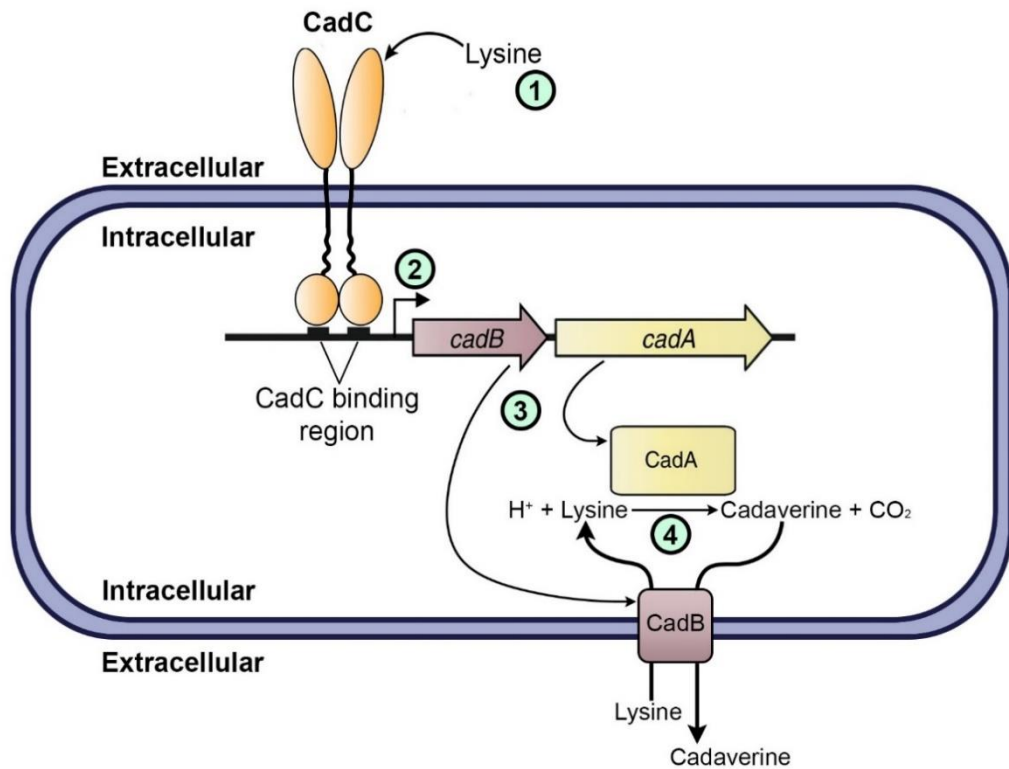


Fig. 4.2

- (i) The operon in Fig. 4.2 is expressed in an inducible manner.

Explain the meaning of 'inducible'.

.....[1]

- (ii) With reference to Fig. 4.2, explain how an activated CadC protein leads to the regulation of extracellular pH in STEC when exposed to acidic conditions.

.....[4]

- (c) STEC produces toxins known as Shiga-toxins which prevent protein synthesis in the host cell and results in cell death. When infected, patients require both anti-toxin and antibiotic treatment.

Suggest why a combination of anti-toxin and antibiotic treatment can more effectively treat STEC-infected patients.

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[Total: 9]

Question 5 starts on page 14

- 5 The Ebola outbreak in West Africa is the world's deadliest to date and the World Health Organization (WHO) declared an international health emergency in response. According to the UN, 1013 people have died in Guinea, Liberia, Sierra Leone and Nigeria in 2014.

Fig. 5.1 below shows some of the main structural features of the Ebola virus.

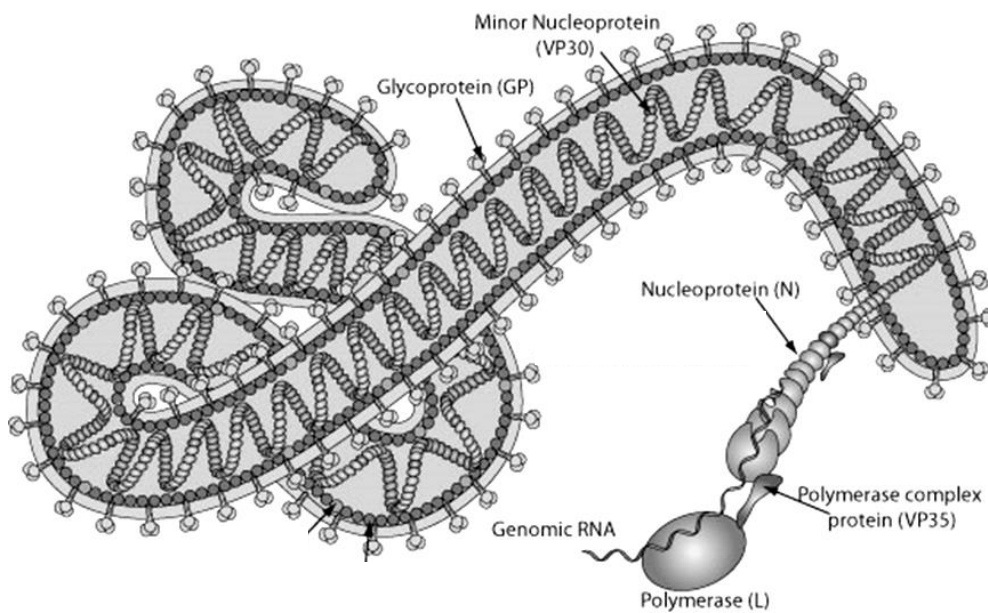


Fig. 5.1

- (a) The Ebola virus (EBOV) is similar to the influenza virus in many ways. For example, EBOV infects the host cell in the same way as influenza virus does.

Using biological knowledge of influenza virus, suggest how EBOV may infect a host cell.

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- (b)** The genome of EBOV contains one molecule of linear, negative sense RNA which is 19kb long. The genome encodes seven proteins, which include nucleoprotein (N), VP30, glycoprotein (GP) and L protein. In particular, L protein is directly involved in genome replication.

Describe the role of L protein.

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- (c)** Despite Ebola being a highly virulent disease in West Africa, suggest why there is little progress in drug development.

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[Total: 9]

6 Mitochondrial DNA (mtDNA) is a double-stranded molecule that consists of approximately 33,000 nucleotides.

- (a) The two strands of mtDNA differ in their base composition, thus allowing both strands to be separated to give a heavy and a light strand by density centrifugation.

Suggest how the heavy strand differs from the light strand.

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[1]

The replication of mtDNA involves the Twinkle protein, which has both helicase and primase activity.

In humans, the Twinkle protein is encoded by the *TWINK* gene that is located in the long arm of chromosome 10.

- (b) Fig. 6.1 is a simplified diagram of a human cell, which shows some key structures typically present.

On Fig. 6.1, use a labelling line and

- (i) the letter **P** to label the precise location where transcription of the *TWINK* gene occurs. [1]
 (ii) the letter **Q** to label the precise location where the Twinkle protein carries out its function. [1]

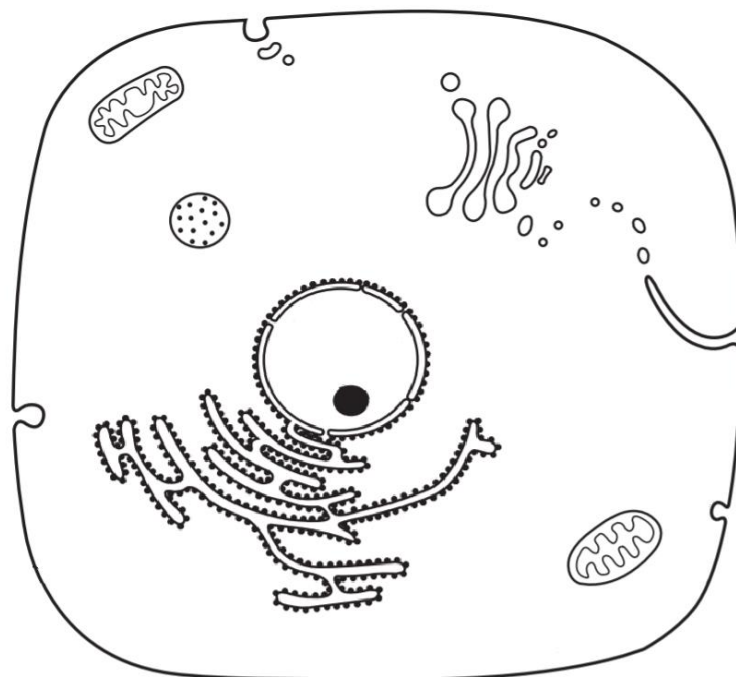


Fig. 6.1

- (c) Explain the functions of the Twinkle protein during the replication of mtDNA.

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.....[3]

- (d) The *TWINK* gene is also used to make the Twinky protein. While the function of this protein is unknown, some research suggest that it functions similarly to the Twinkle protein.

Explain how the single *TWINK* gene can be used to make two proteins.

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.....[2]

- (e) Genes found on mtDNA are often examined to establish evolutionary relationships among organisms. Sequences from the same gene across multiple organisms are aligned to compare changes in this region of DNA.

Suggest why mitochondrial genes are often used to study changes in DNA sequences across organisms.

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[Total: 11]

- 7 George Schull, a botanist at Princeton University, conducted a genetic study of a common weed known as shepherd's purse, *Capsella bursa-pastoris*. He studied the shape of its fruit, which could be heart-shaped or narrow respectively, as shown in Fig. 7.1.



Fig. 7.1

When he crossed a pure-breeding plant with heart-shaped fruit to a pure-breeding plant with narrow fruit, the F1 generation all had heart-shaped fruit. When the F1 generation was self-fertilised, the numbers of F2 generation with the respective shaped fruit were recorded as follows:

heart-shaped fruit	2251
narrow fruit	150

Fig. 7.2 shows the possible biochemical pathway that results in the observation made in the F2 generation.

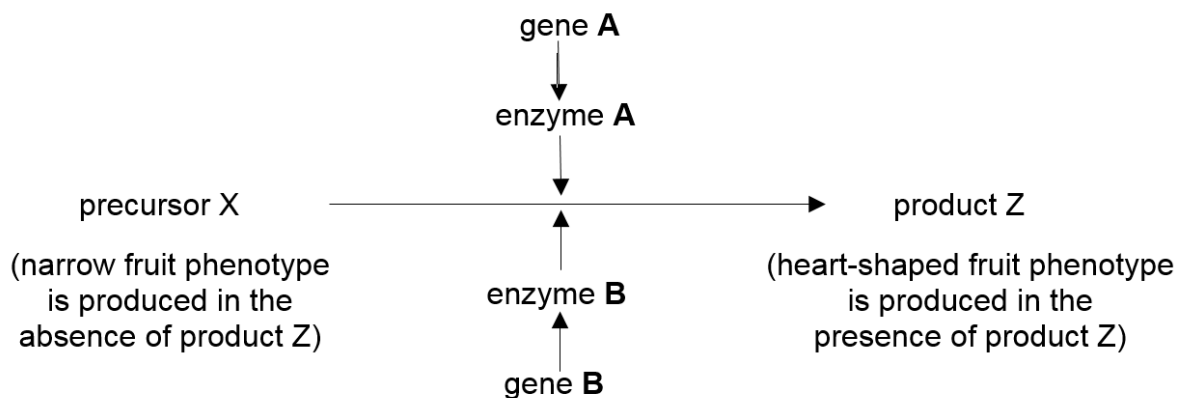


Fig. 7.2

- (a) Explain the type of gene interaction observed in this context.

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- (b) Using the symbols **A**, **a**, **B** and **b**, draw a genetic diagram to explain the results of the F2 generation in the space provided.

[5]

- (c) Describe how *Capsella sp.* plants with heart-shaped fruits can be confirmed to be homozygous dominant in at least one gene locus.

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[Total: 10]

Question 8 starts on page 22

8 Yeast cells carry out respiration to produce ATP for cellular activities.

(a) In a yeast cell, molecules of glucose are phosphorylated at the start of glycolysis.

Suggest why the phosphorylated glucose molecules do not diffuse out of the cell.

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(b) Sometimes, yeast cells will need to carry out respiration in anaerobic conditions to produce ATP.

Explain why the respiration of glucose in anaerobic conditions produces less ATP than in aerobic conditions.

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- (c) One of the by-products of respiration is hydrogen peroxide, which can have harmful effects in yeast cells at high concentrations. Hydrogen peroxide is rapidly broken down to water and oxygen by the enzyme catalase.

A student is asked to carry out an investigation into the effect of hydrogen peroxide concentration on the rate of its breakdown by catalase.

- (i) Outline the key steps in this investigation including how the student can measure the rate of hydrogen peroxide breakdown.

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- (ii) Predict the expected result by sketching a graph in the space below.

Include appropriate labels for the graph axes.

[2]

[Total: 12]

- 9 Fig. 9.1 shows the nucleotide base sequence of a length of DNA from the gene coding for a cytochrome from a modern human, *Homo sapiens*, and the corresponding base sequences from a Neanderthal and four other primate species.

Neanderthals were human or human-like animals that became extinct about 30 000 years ago and are known only from fossil remains. Chimpanzee, orangutan and gorilla are species of ape, while the macaque is a type of monkey.

modern human	ATGACCCCAATACGC AAAATTAAACCCCTAATAAAATTAATTAACCACTCA
Neanderthal	ATGACCCCAATACGC AAAATTAAACCCCTAATAAAATTAATTAACCACTCA
chimpanzee	ATGACCCCAACACGC AAAATTAAACCCCTAATAAAATTAATTAATCACTCA
orangutan	ATGACCTCAACACGTAAAAACAACCCCTAATAAAATTAATCAACCACTCA
gorilla	ATGACCCCTATACGC AAAACTAAACCCCTAACAAAATAATTAACCACTCA
macaque	ATGACTCCAATACGC AAAATCCAACCCCTAACAAAATAATTAATCGCTCC

Fig. 9.1

- (a) Sometimes it is possible to obtain small quantities of DNA from fossils.

Name the technique that is used to provide multiple copies of such DNA molecules for analysis.

.....[1]

- (b) The scientific name for the Neanderthal is considered to be either *Homo sapiens* or *Homo neanderthalensis*.

A scientist considers that *Homo sapiens* is the correct name to use.

State the assumption that the scientist is making and explain why it is difficult to confirm this assumption.

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- (c) Suggest what the data in Fig. 9.1 indicate about the evolutionary relationships between modern humans, Neanderthals and the other primate species.

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- (d) Explain why any conclusions derived from the information in Fig. 9.1 need to be treated with caution.

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[Total: 10]

10 (a) Complete Table 10.1 to show the disease to which the statements apply.

Put a tick (✓) if the statement applies.

Table. 10.1

statement	disease		
	small pox	sickle cell anaemia	tuberculosis
causative agent is a bacterium			
causative agent is a virus			
is an infectious disease			
is a genetic disease			
is preventable by vaccines			

[3]

(b) In humans, three genes code for the production of antibodies:

- Heavy chain gene, which comprises 50 variable segments, 25 diversity segments and 6 joining segments.
- Kappa light chain gene, which comprises 40 variable segments and 5 joining segments.
- Lambda light chain gene, which comprises 30 variable segments and 4 joining segments.

A student calculated the total number of possible antibodies that vary at their antigen-binding site using the equation below:

$$(50 \times 25 \times 6) \times (40 \times 5) \times (30 \times 4) = 1.8 \times 10^8$$

(i) Explain why the student's method of calculation is incorrect **and** show the correct answer with working.

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total number of antibodies [4]

- (ii) Suggest why the total number that you have calculated in (b)(i) is an underestimate of the actual number that may be produced in the human body.

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.....[2]

Fig. 10.1 shows a naïve B cell and a plasma cell. Plasma cells are derived from a naïve B cell.

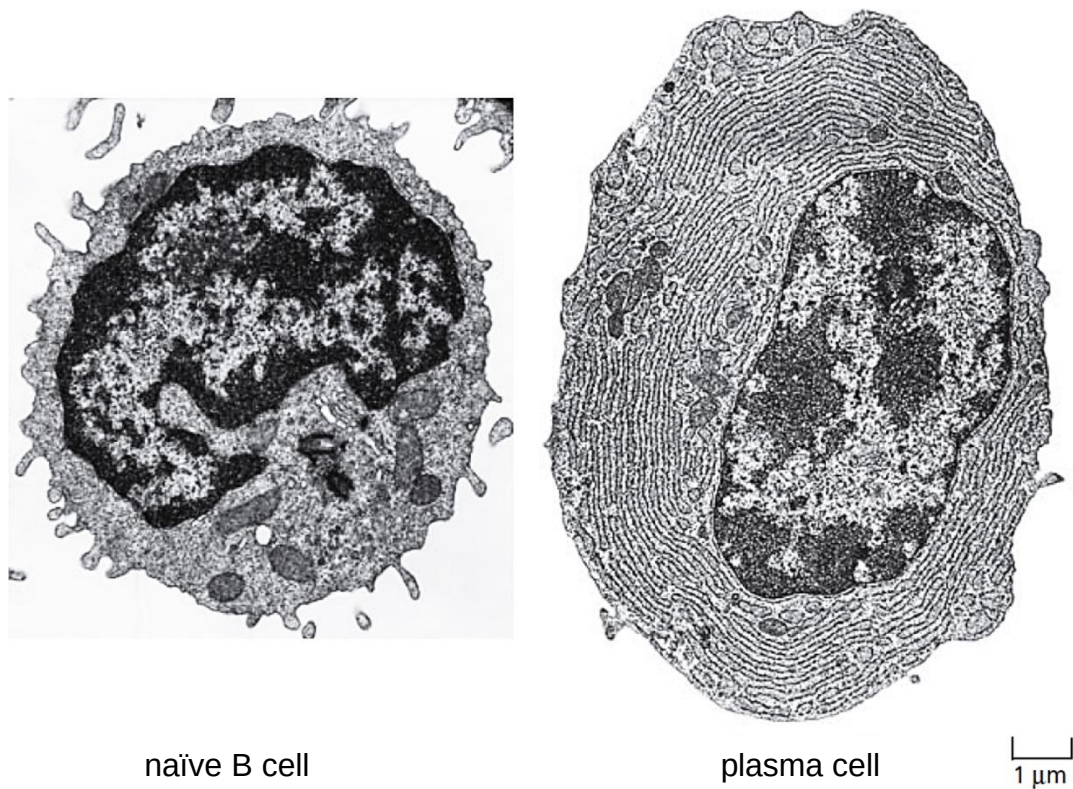


Fig. 10.1

- (c) Apart from the shape and size of the cells, state one other structural difference between a naïve B cell and a plasma cell **and** explain why this difference exists.

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.....[2]

[Total: 11]

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